Théo Riou

Curriculum Vitae

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Personal Information

Date of Birth 18th June 2000 in France
Personal Page https://thriou.github.io/

Education

11/2023 – **PhD in Combustion**, *IMFT, Université de Toulouse*, France.
Topic: Numerical simulation of the dynamics of hydrogen combustion in an aircraft injector.
Keywords: Anchoring dynamics, lift-off dynamics, low and high frequency thermoacoustic instabilities, combustion noise.
Supervisors: Laurent Selle, Guillaume Daviller.
Funding: ERC SELECT-H, led by Thierry Schuller.

2022 – 2023 **Master 2 Combustion**, *ENSMA*, Poitiers, France.
- *Turbulent combustion, Combustion in two-phase flows*

2020 – 2023 **Engineering degree**, *ENSMA*, Poitiers, France.
- *Combustion, Mathematics for Mechanics (hyperbolic systems, distribution theory, optimisation etc.), Numerical Scientific Computation, Finite Elements, Statistical Physics, Turbulence, CFD for industry etc.*

2018 – 2020 **"Classe préparatoire aux grandes écoles" MPSI/MP**, *Victor Grignard*, Cherbourg, France.

2018 **"Baccalauréat scientifique"**, *Victor Grignard*, Cherbourg, France.
- *Mention Très Bien (High Distinction)*

Experience

04/2023 – 09/2023 **Research internship**, *CEA, Paris Saclay*, France.
- *Simulation of a centrifugal pump using Neptune CFD to validate two-phase flow conditions.*

06/2022 – 08/2022 **Research internship**, *University of Leeds*, England.
- *Calibration of a turbulence model for viscoelastic fluids to reduce drag in pipes using OpenFoam.*

Scientific contributions

[1] Riou *et al.* Numerical simulation of flame lift-off dynamics in a hydrogen dual-swirl burner, Proc. Combust. Inst. (2026) (accepted)

[2] McDermott *et al.* An open-source anisotropic $k-\epsilon-v^2-f$ model for turbulent viscoelastic duct flows, Phys. Fluids (2023), doi: 10.1063/5.0159668

Talks and Poster sessions

27–31/07/2026 **Poster**, *International Symposium on Combustion*, Kyoto, Japan.
27-28/11/2025 **Talk**, *Journées thématiques combustion hydrogène*, Saclay, France.
05/06/2025 **Poster**, *Journée des doctorants*, Toulouse, France.
07-11/04/2025 **Poster**, *European Combustion Meeting*, Edinburgh, Scotland.
26-29/02/2024 **Talk**, *Hydrogen week*, Toulouse, France.

Grants and Honors

2025 **HPC allocation**, *EuroHPC*

- *Together with two colleagues, secured a significant European HPC computing grant based on a proposal (70 million CPU hours).*

Responsibilities

2026 **International seminar organisation**, *Hylon Days*, IMFT, France.

2025-2026 **Weekly seminars organisation**, IMFT, France.

2024 **International seminar organisation**, *Hydrogen week: towards high pressure experiments and simulations*, IMFT, France.

Teaching

2026 **Internship supervision (M1)**, *ENSEEIHT*, Toulouse, France.

- *Internship on spectral analysis and modal decomposition for studying thermoacoustic instabilities in a hydrogen burner.*

2024-2025 **Teaching**, *ENSEEIHT*, Toulouse, France.

- *M1 turbulence practical works: hot wire, Particle Image Velocimetry*
- *M1 project: modal decomposition in order to study combustion noise*

Skills

Language French (native), English (B2, TOEIC : 890), Spanish (A2), German (A1).

Computer

- **Simulation software:** AVBP, Neptune CFD, Cantera, OpenFoam
- **Meshing:** Centaur, Salome
- **CAD:** Catia
- **Postprocessing:** Paraview
- **Other:** Python, Antares, Fortran, Linux, GeoGebra

Area Of Interest

Music piano (>19 years), national piano competitions, musical theory

Maths amateur research, reading

Philosophy/theology reading, learning programs (collège des Bernardins)

Sport self-guided hiking (mountains, GR etc.)

REFERENCES

- Laurent Selle e-mail: laurent.selle@imft.fr